

OUTPUT EXPERIMENT-7

```
Problems  Output  Debug Console  Terminal  Ports  +
Final Roll Numbers: 24BAD054
PS C:\Users\kamal\Downloads> & C:/Users/kamal/AppData/Local/Programs/
Python/Python312/python.exe c:/Users/kamal/Downloads/scenario_1_kmean
s.py

--- SCENARIO 1: K-MEANS CLUSTERING ---
Elbow Curve saved as 'kmeans_elbow_curve.png'
Optimal K selected: 5
Inertia (WCSS): 65.57
Silhouette Score (K-Means): 0.5547
K-Means Clusters visualization saved as 'kmeans_clusters.png'

--- Cluster Interpretation ---
  KMeans_Label  ...      Age
0              0  ...  42.716049
1              1  ...  32.692308
2              2  ...  25.272727
3              3  ...  41.114286
4              4  ...  45.217391

[5 rows x 4 columns]

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```
PS C:\Users\kamal\Downloads> & C:/Users/kamal/AppData/Local/Programs/Python/Python312/python.exe c:/Users/kamal/Downloads/scenario_2_gmm.py
```

--- SCENARIO 2: GAUSSIAN MIXTURE MODELS ---

GMM Model Selection saved as 'gmm_aic_bic.png'

Log-Likelihood: -452.50

AIC: 963.00

BIC: 1058.65

Silhouette Score (GMM): 0.5537

GMM Clusters visualization saved as 'gmm_clusters.png'

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```
PS C:\Users\kamal\Downloads> 
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